```
    x+y-z & = 1\ntag\\
    x-y+z & = 1\notag\\
    \intertext{and by hypothesis}
    x+y+z & =1\tag{H}
  \end{align*}
```

which gives the following output

Thus x, y and z satisfy the equations

$$\begin{aligned} x+y-z &= 1 \\ x-y+z &= 1 \end{aligned}$$

and by hypothesis

$$x+y+z = 1 \tag{H}$$

What about `split` and `aligned`? As we have seen, these can be used only within some other equation structure. The numbering or the lack of it is determined by this parent structure. Thus

```
\begin{equation}
  \begin{split}
    (a+b)^2 & = (a+b)(a+b)\\
            & = a^2+ab+ba+b^2\\
            & = a^2+2ab+b^2
  \end{split}
\end{equation}
```

gives

$$\begin{aligned} (a+b)^2 &= (a+b)(a+b) \\ &= a^2+ab+ba+b^2 \\ &= a^2+2ab+b^2 \end{aligned} \tag{VIII.5}$$

VIII.4. MATHEMATICS MISCELLANY

There are more things Mathematics than just equations. Let us look at how LaTeX and in particular, the amsmath package deals with them.

VIII.4.1. Matrices

Matrices are by definition numbers or mathematical expressions arranged in rows and columns. The amsmath has several environments for producing such arrays. For example

The system of equations

$$\begin{aligned} x+y-z &= 1\\ x-y+z &= 1\\ x+y+z &= 1 \end{aligned}$$

can be written in matrix terms as

$$\begin{pmatrix} 1 & 1 & -1\\ 1 & -1 & 1\\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x\\ y\\ z \end{pmatrix} = \begin{pmatrix} 1\\ 1\\ 1 \end{pmatrix}.$$

Here, the matrix $\begin{pmatrix} 1 & 1 & -1\\ 1 & -1 & 1\\ 1 & 1 & 1 \end{pmatrix}$ is invertible.

is produced by

```
The system of equations
\begin{align*}
  x+y-z & = 1\\
  x-y+z & = 1\\
  x+y+z & = 1
\end{align*}
can be written in matrix terms as
\begin{equation*}
  \begin{pmatrix}
    1 &  1 & -1\\
    1 & -1 &  1\\
    1 &  1 &  1
  \end{pmatrix}
  \begin{pmatrix}
    x\\
    y\\
    z
  \end{pmatrix}
  =
  \begin{pmatrix}
    1\\
    1\\
    1
  \end{pmatrix}.
\end{equation*}
Here, the matrix
$\begin{pmatrix}
  1 &  1 & -1\\
  1 & -1 &  1\\
  1 &  1 &  1
\end{pmatrix}$
is invertible.
```

Note that the environment `pmatrix` can be used within in-text mathematics or in displayed math. Why the `p`? There is indeed an environment `matrix` (without a `p`) but it

produces an array *without* the enclosing parentheses (try it). If you want the array to be enclosed within *square brackets*, use `bmatrix` instead of `pmatrix`. Thus

Some mathematicians write matrices within parentheses as in $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ while others prefer square brackets as in $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

is produced by

```
Some mathematicians write matrices within parentheses as in
$
\begin{pmatrix}
  a & b\\
  c & d
\end{pmatrix}
$
while others prefer square brackets as in
$
\begin{bmatrix}
  a & b\\
  c & d
\end{bmatrix}
$
```

There is also a `vmatrix` environment, which is usually used for determinants as in

The determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is defined by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

which is obtained from the input

```
The determinant
$
\begin{vmatrix}
  a & b\\
  c & d
\end{vmatrix}
$
is defined by
\begin{equation*}
  \begin{vmatrix}
    a & b\\
    c & d
  \end{vmatrix}
  =ad -bc
\end{equation*}
```

There is a variant `Vmatrix` which encloses the array in double lines. Finally, we have a `Bmatrix` environment which produces an array enclosed within braces { }.

A row of dots in a matrix can be produced by the command `\hdotsfour`. it should be used with an argument specifying the number of columns to be spanned. For example, to get

A general $m \times n$ matrix is of the form

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

we type

```
A general $m\times n$ matrix is of the form
\begin{equation*}
  \begin{pmatrix}
    a_{11} & a_{12} & \dots & a_{1n}\\
    a_{21} & a_{22} & \dots & a_{2n}\\
    \hdotsfor{4}\\
    a_{m1} & a_{m2} & \dots & a_{mn}
  \end{pmatrix}
\end{equation*}
```

The command `\hdotsfor` has also an optional argument to specify the spacing of dots. Thus in the above example, if we use `\hdotsfor[2]{4}`, then the space between the dots is doubled as in

A general $m \times n$ matrix is of the form

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

VIII.4.2. Dots

In the above example, we used the command `\dots` to produce a row of three dots. This can be used in other contexts also. For example,

```
Consider a finite sequence $X_1,X_2,\dots$, its sum $X_1+X_2+\dots$
and product $X_1X_2\dots$.
```

gives

Consider a finite sequence $X_1, X_2, \dots$, its sum $X_1 + X_2 + \dots$ and product $X_1X_2 \dots$.

Here the dots in all the three contexts are along the "baseline" of the text. Isn't it better to typeset this as

Consider a finite sequence $X_1, X_2, \dots$, its sum $X_1 + X_2 + \cdots$ and product $X_1X_2 \cdots$.

with *raised* dots for addition and multiplication? The above text is typeset by the input

```
Consider a finite sequence $X_1,X_2,\dotsc$, its sum $X_1+X_2+\dotsb$
and product $X_1X_2\dotsm$.
```

Here `\dotsc` stands for dots to be used with commas, `\dotsb` for dots with binary operations (or relations) and `\dotsm` for multiplication dots. There is also a `\dotsi` for dots with integrals as in

$$\int_{A_1}\int_{A_2}\cdots\int_{A_n} f$$

VIII.4.3. Delimiters

How do we produce something like

Since $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$, the matrix $\begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix}$ is not invertible.

Here the 'small' in-text matrices are produced by the environment `smallmatrix`. This environment *does not* provide the enclosing *delimiters* ( ) or — — which we must supply as in

```
$
\left|\begin{smallmatrix}
        a & h & g\\
        h & b & f\\
        g & f & c
      \end{smallmatrix}\right|
=0
$,
the matrix
$
\left(\begin{smallmatrix}
        a & h & g\\
        h & b & f\\
        g & f & c
      \end{smallmatrix}\right)
$
is not invertible.
```

Why the `\left|...\right|` and `\left{...\right?` These commands `\left` and `\right` enlarge the *delimiter* following them to the size of the enclosed material. To see their effect, try typesetting the above example without these commands. The list of symbols at the end of the chapter gives a list of delimiters that are available off the shelf.

One interesting point about the `\left` and `\right` pair is that, though every `\left` should be matched to a `\right`, the *delimiters* to which they apply need not match. In particular we can produce a single large delimiter produced by `\left` or `\right` by matching it with a matching command followed by a period. For example,

$$\left.\begin{aligned} u_x &= v_y \\ u_y &= -v_x \end{aligned}\right\} \quad \text{Cauchy-Riemann Equations}$$

is produced by

```
\begin{equation*}
  \left.
    \begin{aligned}
      u_x &= v_y\\
      u_y &= -v_x
    \end{aligned}
  \right\}
  \quad\text{Cauchy-Riemann Equations}
\end{equation*}
```

There are instances where the delimiters produced by `\left` and `\right` are too small or too large. For example,

```
\begin{equation*}
  (x+y)^2-(x-y)^2=\left((x+y)+(x-y)\right)\left((x+y)-(x-y)\right)=4xy
\end{equation*}
```

gives

$$(x+y)^2-(x-y)^2=((x+y)+(x-y))\,((x+y)-(x-y))=4xy$$

where the parentheses are all of the same size. But it may be better to make the outer ones a little larger to make the nesting visually apparent, as in

$$(x+y)^2-(x-y)^2=\Big((x+y)+(x-y)\Big)\Big((x+y)-(x-y)\Big)=4xy$$

This is produced using the commands `\bigl` and `\bigr` before the outer parentheses as shown below:

```
\begin{equation*}
  (x+y)^2-(x-y)^2=\bigl((x+y)+(x-y)\bigr)\bigl((x+y)-(x-y)\bigr)=4xy
\end{equation*}
```

Apart from `\bigl` and `\bigr` there are `\Bigl`, `\biggl` and `\Biggl` commands (and their `r` counterparts) which (in order) produce delimiters of increasing size. (Experiment with them to get a feel for their sizes.)

As another example, look at

For n-tuples of complex numbers $(x_1,x_2,\dots,x_n)$ and $(y_1,y_2,\dots,y_n)$ of complex numbers

$$\left(\sum_{k=1}^{n}|x_ky_k|\right)^2\le\left(\sum_{k=1}^{n}|x_k|\right)\left(\sum_{k=1}^{n}|y_k|\right)$$

which is produced by

```
For $n$-tuples of complex numbers $(x_1,x_2,\dotsc,x_n)$ and
$(y_1,y_2,\dotsc,y_n)$ of complex numbers
\begin{equation*}
  \left(\sum_{k=1}^n|x_ky_k|\right)^2\le
        \left(\sum_{k=1}^{n}|x_k|\right)\left(\sum_{k=1}^{n}|y_k|\right)
\end{equation*}
```

Does not the output below look better?

For n-tuples of complex numbers $(x_1, x_2, \dots, x_n)$ and $(y_1, y_2, \dots, y_n)$ of complex numbers

$$\Big(\sum_{k=1}^{n} |x_k y_k|\Big)^2 \le \Big(\sum_{k=1}^{n} |x_k|\Big)\Big(\sum_{k=1}^{n} |y_k|\Big)$$

This one is produced by

```
For $n$-tuples of complex numbers $(x_1,x_2,\dotsc,x_n)$ and
$(y_1,y_2,\dotsc,y_n)$ of complex numbers
\begin{equation*}
  \biggl(\sum_{k=1}^n|x_ky_k|\biggr)^2\le
    \biggl(\sum_{k=1}^{n}|x_k|\biggr)\biggl(\sum_{k=1}^{n}|y_k|\biggr)
\end{equation*}
```

Here the trouble is that the delimiters produced by `\left` and `\right` are a bit too large.

VIII.4.4. Putting one over another

Look at the following text

From the binomial theorem, it easily follows that if n is an even number, then

$$1 - \binom{n}{1}\frac{1}{2} + \binom{n}{2}\frac{1}{2^2} - \cdots - \binom{n}{n-1}\frac{1}{2^{n-1}} = 0$$

We have fractions like $\frac{1}{2^{n-1}}$ and *binomial coefficients* like $\binom{n}{2}$ here and the common feature of both is that they have one mathematical expression over another.

Fractions are produced by the `\frac` command which takes two arguments, the numerator followed by the denominator and the binomial coefficients are produced by the `\binom` command which also takes two arguments, the 'top' expression followed by the 'bottom' one. Thus the the input for the above example is

```
From the binomial theorem, it easily follows that if $n$ is an even
number, then
\begin{equation*}
  1-\binom{n}{1}\frac{1}{2}+\binom{n}{2}\frac{1}{2^2}-\dotsb
   -\binom{n}{n-1}\frac{1}{2^{n-1}}=0
\end{equation*}
```

You can see from the first paragraph above that the *size* of the outputs of `\frac` and `\binom` are smaller in text than in display. This default behavior has to be modified sometimes for nicer looking output. For example, consider the following output

Since (x_n) converges to 0, there exists a positive integer p such that

$$|x_n| < \frac{1}{2} \quad \text{for all } n \ge p$$

Would not it be nicer to make the fraction smaller and typeset this as

Since (x_n) converges to 0, there exists a positive integer p such that

$$|x_n| < \tfrac{1}{2} \quad \text{for all } n \ge p$$

The second output is produced by the input

```
Since $(x_n)$ converges to $0$, there exists a positive integer $p$
such that
\begin{equation*}
  |x_n|<\tfrac{1}{2}\quad\text{for all $n\ge p$}
\end{equation*}
```

Note the use of the command `\tfrac` to produce a smaller fraction. (The first output is produced by the usual `\frac` command.)

There is also command `\dfrac` to produce a display style (larger size) fraction in text. Thus the sentence after the first example in this (sub)section can be typeset as

> We have fractions like $\dfrac{1}{2^{n-1}}$ and ...

by the input

```
We have fractions like $\dfrac{1}{2^{n-1}}$ and ...
```

As can be guessed, the original output was produced by `\frac`. Similarly, there are commands `\dbinom` (to produce display style binomial coefficients) and `\tbinom` (to produce text style binomial coefficients).

There is also a `\genfrac` command which can be used to produce custom fractions. To use it, we will have to specify six things

1. The left delimiter to be used—note that { must be specified as `\{`
2. The right delimiter—again, } to be specified as `\}`
3. The thickness of the horizontal line between the top expression and the bottom expression. If it is not specified, then it defaults to the 'normal' thickness. If it is set as `0pt` then there will be no such line at all in the output.
4. The size of the output—this is specified as an integer 0, 1, 2 or 3, greater values corresponding to *smaller* sizes. (Technically these values correspond to `\displaystyle`, `\textstyle`, `\scriptstyle` and `\scriptscriptstyle`.)
5. The top expression
6. The bottom expression

Thus instead of `\tfrac{1}{2}` we can also use `\genfrac{}{}{}{1}{1}{2}` and instead of `\dbinom{n}{r}`, we can also use `\genfrac{(}{)}{0pt}{0}{1}{2}` (but there is hardly any reason for doing so). More seriously, suppose we want to produce $\{{ij \atop k}\}$ and $[{ij \atop k}]$ as in

> The Christoffel symbol $\{{ij \atop k}\}$ of the second kind is related to the Christoffel symbol $[{ij \atop k}]$ of the first kind by the equation
> $$\left\{{ij \atop k}\right\} = g^{k1}\left[{ij \atop 1}\right] + g^{k2}\left[{ij \atop 2}\right]$$

This can be done by the input

```
The Christoffel symbol $\{{ij \atop k}\}$ of the second kind is related to the Christoffel symbol $[{ij \atop k}]$ of the first kind by the equation
$$\left\{{ij \atop k}\right\} = g^{k1}\left[{ij \atop 1}\right] + g^{k2}\left[{ij \atop 2}\right]$$
```

If such expressions are frequent in the document, it would be better to define 'newcommands' for them and use them instead of `\genfrac` every time as in the following input (which produces the same output as above).

```
\newcommand{\chsfk}[2]{\genfrac{[}{]}{0pt}{}{#1}{#2}}
\newcommand{\chssk}[2]{\genfrac{\{}{\}}{0pt}{}{#1}{#2}}
The Christoffel symbol $\genfrac{\{}{\}}{0pt}{}{ij}{k}$ of the second
kind is related to the Christoffel symbol $\genfrac{[}{]}{0pt}{}{ij}{k}$
of the first kind by the equation
\begin{equation*}
  \chssk{ij}{k}=g^{k1}\chsfk{ij}{1}+g^{k2}\chsfk{ij}{2}
\end{equation*}
```

While on the topic of fractions, we should also mention the `\cfrac` command used to typeset continued fractions. For example, to get

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cdots}}}$$

simply type

```
\begin{equation*}
  \frac{4}{\pi}=1+\cfrac{1^2}{2+
                  \cfrac{3^2}{2+
                   \cfrac{5^2}{2+\dotsb}}}
\end{equation*}
```

Some mathematicians would like to write the above equation as

$$\frac{4}{\pi} = 1 + \frac{1^2}{2} + \frac{3^2}{2} + \frac{5^2}{2} + \cdots$$

There is no ready-to-use command to produce this, but we can define one as follows

```
\newcommand{\cfplus}{\mathbin{\genfrac{}{}{0pt}{}{}{+}}}
\begin{equation*}
  \frac{4}{\pi}
    =1+\frac{1^2}{2}\cfplus\frac{3^2}{2}\cfplus\frac{5^2}{2}\cfplus\dotsb
\end{equation*}
```

VIII.4.5. Affixing symbols—over or under

The table at the end of this chapter gives various math mode *accents* such as `$\hat{a}$` to produce $\hat{a}$ and `$\dot{a}$` to produce $\dot{a}$. But what if one needs $\overset{\circ}{a}$ or $\underset{\circ}{a}$? The commands `\overset` and `\underset` come to the rescue. Thus `$\overset{\circ}{a}$` produces $\overset{\circ}{a}$ and `$\underset{\circ}{a}$` produces $\underset{\circ}{a}$.

Basic LaTeX provides the commands `\overrightarrow` and `\overleftarrow` also to put (extensible) arrows over symbols, as can be seen from the table. The amsmath package also provides the commands `\underrightarrow` and `\underleftarrow` to put (extensible) arrows *below* mathematical expressions.

Speaking of arrows, amsmath provides the commands `\xrightarrow` and `\xleftarrow` which produces arrows which can accommodate long texts as superscripts or subscripts. Thus we can produce

Thus we see that

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

is a short exact sequence

from the input

```
Thus we see that
\begin{equation*}
  0\xrightarrow{} A\xrightarrow{f}
                  B\xrightarrow{g}
                  C\xrightarrow{} 0
\end{equation*}
is a short exact sequence
```

Note how the *mandatory* arguments of the first and last arrows are left empty to produce arrows with no superscripts. These commands also allow an *optional* argument (to be typed inside *square brackets*), which can be used to produce subscripts. For example

```
Thus we get
\begin{equation*}
  0\xrightarrow{} A\xrightarrow[\text{monic}]{f}
                  B\xrightarrow[\text{epi}]{g}
                  C\xrightarrow{} 0
\end{equation*}
```

gives

Thus we get

$$0 \to A \xrightarrow[\text{monic}]{f} B \xrightarrow[\text{epi}]{g} C \to 0$$

By the way, would not it be nicer to make the two middle arrows the same width? This can be done by changing the command for the third arrow (the one from `B`) as shown below

```
Thus we get
\begin{equation*}
  0\xrightarrow{} A\xrightarrow[\text{monic}]{f}
                  B\xrightarrow[\hspace{7pt}\text{epi}\hspace{7pt}]{g}
                  C\xrightarrow{}0
\end{equation*}
```

This gives

Thus we get

$$0 \to A \xrightarrow[\text{monic}]{f} B \xrightarrow[\text{epi}]{g} C \to 0$$

where the lengths of the two arrows are *almost* the same. There are indeed ways to make the lengths *exactly* the same, but we will talk about it in another chapter.

Mathematical symbols are also attached as *limits* to such *large operators* as sum ($\sum$), product ($\prod$) set union ($\bigcup$), set intersection ($\bigcap$) and so on. The limits are input as subscripts or superscripts, but their *positioning* in the output is different in text and display. For example, the input

```
Euler not only proved that the series
$\sum_{n=1}^\infty\frac{1}{n^2}$ converges, but also that
\begin{equation*}
  \sum_{n=1}^\infty\frac{1}{n^2}=\frac{\pi^2}{6}
\end{equation*}
```

gives the output

Euler not only proved that the series $\sum_{n=1}^\infty \frac{1}{n^2}$ converges, but also that

$$\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$$

Note that in display, the sum symbol is larger and the limits are put at the bottom and top (instead of at the sides,which is usually the case for subscripts and superscripts). If you want the *same* type of symbol (size, limits and all) in text also, simply change the line

```
$\sum_{n=1}^\infty\frac{1}{n^2}$
```

to

```
$\displaystyle\sum_{n=1}^\infty\frac{1}{n^2}$
```

and you will get

Euler not only proved that the series $\displaystyle\sum_{n=1}^\infty \frac{1}{n^2}$ converges, but also that

$$\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$$

(Note that this also changes the size of the fraction. What would you do to keep it small?) On the other hand, to make the displayed operator the same as in the text, add the command `\textstyle` before the `\sum` within the equation.

What if you only want to change the *position of the limits* but not the size of the operator in text? Then change the command `$\sum_{n=1}^\infty \frac{1}{n^2}$` to `$\sum_\limits{n=1}^\infty\frac{1}{n^2}$` and this will produce the output given below.

Euler not only proved that the series $\sum\limits_{n=1}^\infty \frac{1}{n^2}$ converges, but also that

$$\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$$

On the other hand, if you want side-set limits in display type `\nolimits` after the `\sum` within the equation as in

```
Euler not only proved that the series
$\sum_{n=1}^\infty\frac{1}{n^2}$ converges, but also that
\begin{equation*}
  \sum\nolimits_{n=1}^\infty\frac{1}{n^2}=\frac{\pi^2}{6}
\end{equation*}
```

which gives

Euler not only proved that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, but also that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

All these are true for other operators classified as "Variable-sized symbols",except integrals. Though the integral symbol in display is larger, the position of the limits in both text and display is on the side as can be seen from the output below

Thus $\lim\limits_{x\to\infty}\int_0^x \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$ and so by definition,

$$\int_0^{\infty} \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$$

which is produced by

```
Thus
$\lim\limits_{x\to\infty}\int_0^x\frac{\sin x}{x}\,\mathrm{d}x
      =\frac{\pi}{2}$
and so by definition,
\begin{equation*}
  \int_0^\infty\frac{\sin x}{x}\,\mathrm{d}x=\frac{\pi}{2}
\end{equation*}
```

If you want the limits to be above and below the integral sign, just add the command `\limits` immediately after the `\int` command. Thus

```
Thus
$\lim\limits_{x\to\infty}\int_0^x\frac{\sin x}{x}\,\mathrm{d}x
      =\frac{\pi}{2}$
and so by definition,
\begin{equation*}
  \int\limits_0^\infty\frac{\sin x}{x}\,\mathrm{d}x=\frac{\pi}{2}
\end{equation*}
```

gives

Thus $\lim\limits_{x\to\infty}\int_0^x \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$ and so by definition,

$$\int\limits_0^{\infty} \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$$

Now how do we typeset something like

$$p_k(x) = \prod_{\substack{i=1 \\ i\neq k}}^{n} \left(\frac{x - t_i}{t_k - t_i}\right)$$

where we have two lines of subscripts for $\prod$? There is a command `\substack` which will do the trick. The above output is obtained from